

Automated Liquid Handler from a 3D Printer

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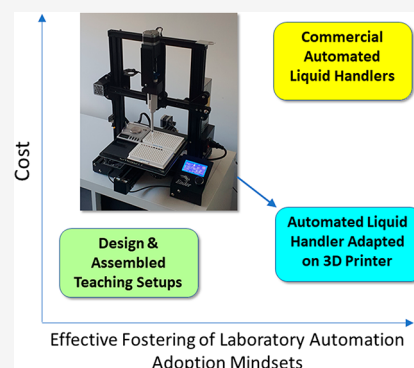


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ABSTRACT: Efforts to encourage laboratory automation through student access to and usage of commercial equipment are hampered by prohibitive acquisition costs. Here, an automated liquid handler using manual pipettes was developed to operate on a commercial hobbyist 3D printer at a cost of US \$325. Arising from an iterative design process that ensured sufficient strength and rigidity, the adapter assembly, when fabricated by using 3D printing, was found to operate as it was intended. Tests conducted found the system to be sufficiently stable and accurate in its positioning, while pipetting accuracy was not affected. A survey following a usage exercise with the operational system was conducted as part of an overall effort to encourage students to overcome their disciplinary boundaries.



KEYWORDS: Laboratory Instruction, High School/Introductory Chemistry, First-Year Undergraduate/General, 3D Printing

INTRODUCTION

Chemistry and biology experiments are essential for the identification and synthesis of novel organic and nonorganic materials and the verification of their performance characteristics. Traditionally, these experiments have been conducted by skilled persons in the laboratory. There is growing interest to utilize higher levels of automation in the chemistry and biology laboratory to overcome this requirement.^{1,2} The transition to this mode of operation, however, requires the involvement of personnel that are, in the first instance, comfortable with adopting automation and robotics technology in the laboratory. The Technology Acceptance Model (TAM), which is arguably the most widely tested model across many fields,^{3,4} has consistently confirmed that a user's overall attitude toward embracing any technology is determined by two factors: its perceived usefulness and its perceived ease of usage. The formative learning periods in high school and university logically present the most opportune times to nurture these perceptions. There have been reported attempts to introduce aspects of automation into experiments that are routinely conducted in the teaching laboratory.^{5–7} While these setups are helpful to some extent, the availability of de facto instrumentation (those that are used in research and industrial laboratories) for students to operate and explore on should have a stronger impact in fostering the desired technology acceptance persuasions more positively. In addition, this mode abets efforts to improve employability upon graduation.⁸ The unfortunate reality, however, is that such equipment is often costly to acquire. The availability of cost-effective alternative instrumentation, which does not have the full range of

functions or the high precision of the research and industrial versions but maintains similar working principles, may suffice in fulfilling this need.

A highly routine task that is conducted manually and repetitively in chemistry and biochemistry laboratories is aliquoting minute quantities of liquids accurately without the risk of contamination (i.e., pipetting). The high cost of commercial automated liquid handlers (ALHs) is prohibitive even in research and commercial laboratories, and this has driven efforts to develop cheaper alternatives.⁹ The material costs alone to acquire each of these supposedly lower cost machines have been estimated to range around US \$6,000, and it excludes the labor needed to fabricate and assemble the custom designs as well as to program the electronic controllers properly. More rudimentary constructions, while being able to reduce costs further, require a considerable investment in time and effort as well as the technical knowledge needed to assemble and to program their basic operation.^{10,11} Importantly, these assemblies do not have the appearances that would persuade adoption.

3D printing technology encompasses an additive process whereby three-dimensional solid objects are manufactured layer-by-layer from computer-aided design (CAD) drawings.

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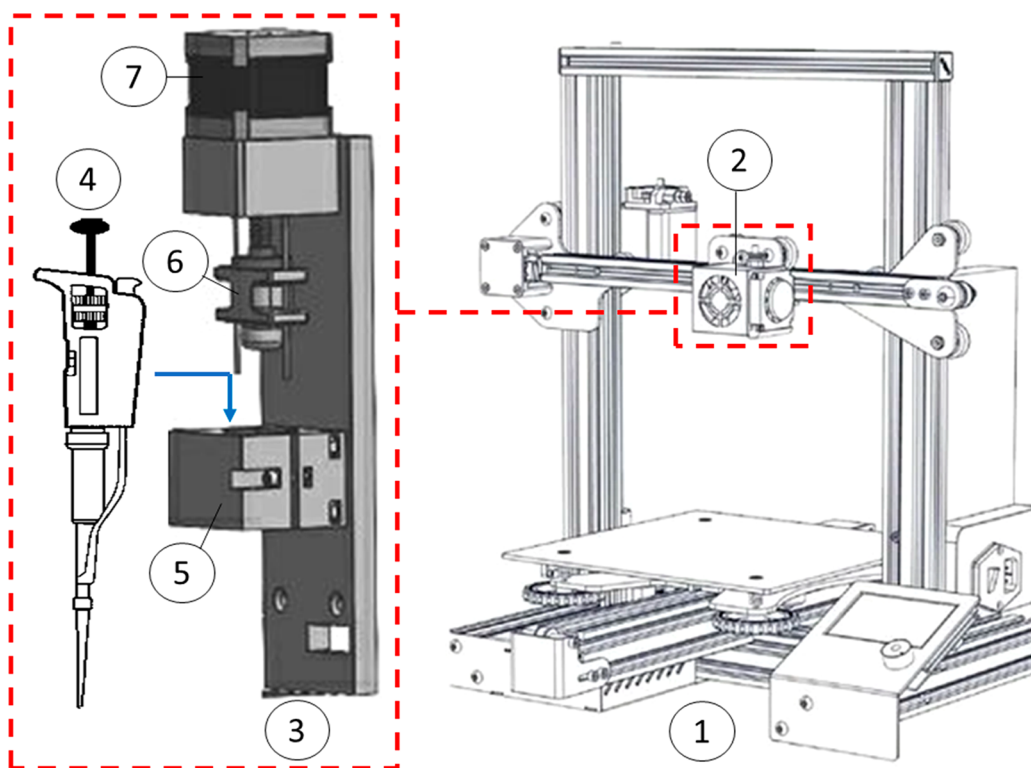


Figure 1. Prototype system (see [Supplementary Video](#)) with a (1) commercial hobbyist 3D printer, in which the (2) hot-end module was replaced by a (3) bespoke pipetting adapter assembly which allowed a (4) standard manual pipette to be attached to the (5) fitting mount. The plunger of the manual pipette was actuated using a (6) ball-screw slider driven by a (7) stepper motor.

Code to apply homing routine

comments after the comma sign

G28 X Y; home the X and Y axes of the printer

G90 ; use absolute positioning for the XYZ axes

G1 X10 F3600; move 10mm to the right of the current location

instruction to set speed

instruction to move in the x direction

code to use linear movement

Figure 2. An illustration of three typical G-code command lines used to control a 3D printer operation. Each line comprises a code and some instructions to achieve specific operations. Comments are often also included to enable an easy understanding.

While there are a variety of fabrication approaches to do this,¹² it is the fused deposition melting (FDM) method that has spawned the prodigious growth in hobbyist 3D printers that are currently priced between US \$200 to \$800 each to acquire. These printers are modular in design, making it possible to readily remove and replace any of the components that make up their whole. In addition, users are able to instruct the printer to carry out various actions: such as setting the position of the printer head, speed of the printer head, extruder temperature, bed temperature, and other parameters via commands written in a universal programming language (G-code).¹³ Every printer is equipped with the necessary firmware (software that runs on the 3D printer's microcontroller) to translate these G-code instructions into low-level hardware

control signals. These 3D printers, because they are manufactured at industrial scales, appear in forms that are close to typical commercial automated laboratory equipment, while being relatively inexpensive.

In this work, a commercial hobbyist 3D printer (Creality Ender 3 Pro) was adapted to operate as an automated liquid handler that uses manual pipets already available in the laboratory. The design process in creating the adapter assembly as well as the method to use a laptop or computer to control the pipetting operation are outlined. Tests to ascertain the positioning stability and accuracy of the system, as well as the effect on pipetting accuracy, were conducted. A survey was also carried out with 9 student participants following a usage exercise with the system.

MATERIALS AND METHODS

System Development

A commercial hobbyist 3D printer (Creality Ender 3 Pro) was adapted to operate as an automated liquid handler by replacing the hot-end module of the printer with a bespoke pipetting adapter assembly (Figure 1). Actuation on the printer is achieved via four stepper motors; three for motion and one to operate the hot-end module. The adaptation is based on using the stepper motor meant for the hot-end module (Creality, 42–40) to instead operate the pipetting adapter assembly. Hence, the functions of the liquid handler remain driven only by the original controller of the printer. A commercial 10–100 μL single channel manual pipet (Precipette, Four E's Scientific) was fitted via a mount on the assembly, such that its plunger was actuated via the movement of a ball-screw slider driven by the stepper motor. The pipetting adapter assembly was iteratively designed using CAD (Solidworks).

A series of G-code commands specific to the Marlin firmware (see examples in Figure 2) were written to operate the system and conduct the verification tests. These can be stored in a micro-SD card which can be read by the 3D printer (through its firmware) or by direct instruction from a personal computer (PC) or laptop through a USB cable connection to the printer. The latter is more convenient when frequent changes in the G-code instructions are required. A free-to-use software (Pronterface) installed on the PC was used to interface with the printer as well as to send the operational G-code instructions to it.

Design Assessment

Every pipetting adapter design version was assessed for both practicality and structural integrity. The latter was achieved using the Finite Element Analysis (FEA) facility available in the CAD software (Solidworks) itself. The process involved discretizing the CAD model into a simulation mesh that optimized the mesh density for accuracy, as well as the solving time. The overall strategy involved coming up with 3 design layouts. Once a design layout was decided on based on practicable considerations (such as the ability to load and remove the manual pipette easily, how snugly the pipette fits onto the adapter, and collision avoidance of the pipette tip with the printer bed), an FEA was conducted and minor changes made to the design (e.g., removal of sharp corners, discontinuities etc.) until suitably low stress and deflection characteristics were attained. The parts for the final pipetting adapter design were then fabricated by using the same 3D printer and assembled for use.

System Performance Assessment

Two tests were carried out. In the first, the automated liquid handler was instructed to transfer 100 μL of distilled water (colored with food dye to identify spillage more readily) from all the wells of a 96-well white microplate (PerkinElmer, OptiPlate-96) to the corresponding wells of a similar recipient microplate. The lack of liquid spillage would indicate the positioning stability and accuracy of the system. For the second test, a series of liquid transfers of 75 and 100 μL (set on the pipet) were made onto a precision weighing scale of 0.001 g resolution to determine the actual volume transferred. Measurements made using the automated as well as manual modes were then used to compare their relative performances.

Survey

The survey was conducted on 9 undergraduate students with a background in either Mechanical or Mechatronics Engineering who had just completed a project to design specific biochemical equipment. The students were collectively briefed in a prior session on the design, construction, and operation of the liquid handling system. They were then organized to attend a session alone or with another participant in which they would use the system via programming the relevant G-code instructions to automatically transfer reagents from randomly prescribed wells of a PCR microplate. At the end of the session, the participants were asked to complete a short survey.

RESULTS AND DISCUSSION

It was intended for the designed adapter parts to be manufactured using poly(lactic acid) (PLA), as it was easy to print with, inexpensive, and has reasonably good overall stiffness and strength. PLA is also not soluble in alcohols and linear hydrocarbons. The finalized design was simulated using FEA to be subjected to a force of 25 N acting on the 4 screws located at the mount that held the pipet to the assembly. The Von Mises stress and deflection (see Figure 3) distributions

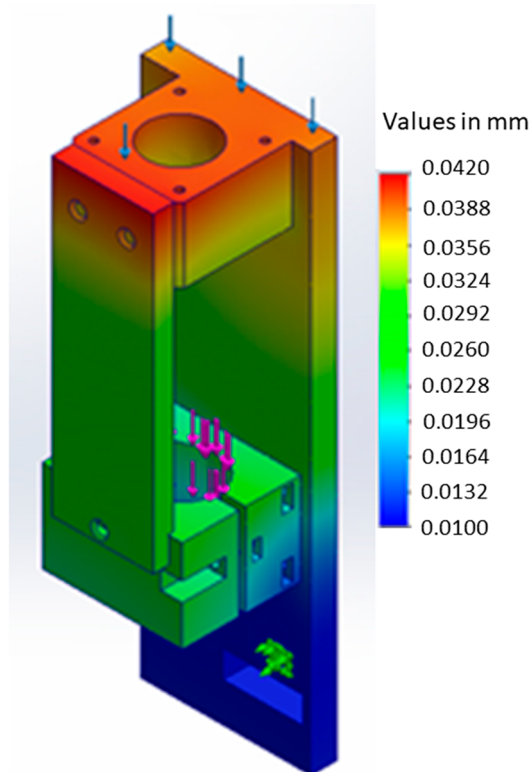


Figure 3. Expected deflection distribution of the finalized pipet adapter design obtained under a simulated loading of 25 N applied on it. The maximum value did not exceed 0.042 mm.

showed higher values at specific regions as would be typically expected. However, the highest stress value was found to not exceed 0.96 MPa, which was significantly lower than the yield strength of the material (~ 26 MPa). Similarly, the deflection was also uncovered to be restricted to 0.042 mm, indicating good rigidity under this expected loading force. These findings indicated that the design ought to be able to operate well

should the parts be fabricated by using 3D printing and assembled.

The 3D printer was fitted with a pipetting assembly in place of the hot-end module (which was temporarily removed). It was found to fit snugly in place and appeared to operate in accordance with the G-code instructions sent to the printer by the computer. A simple calibration process was done in which the pipette tip was moved to determine the coordinates of the centers of the 4 corner wells of the microplate. Following this, it should be relatively straightforward to use the G-codes to move the tip to the theoretical center of any well of the microplate. It is possible to avoid repeating the calibration step by creating a cutout shape jig (easily achieved using a router) that is attached on the 3D printer bed for the microplates to be placed in. With each microplate switch, it will be prudent to move the tip to any random well followed by a simple visual check (to confirm that it corresponded to the center of the well). Despite this, it remained necessary to establish whether pipetting operations could be accurately and repeatedly performed using the G-code instructions.

The two 96-well microplates and weighing scale were located on the 3D printer, as shown in Figure 4. In the first test conducted to transfer 100 μL of distilled water from all the wells of a 96-well white microplate to the corresponding wells of a similar recipient microplate, no spillage was found to occur. This indicated the positioning stability and accuracy of the system. A subsequent test was conducted with a sequence

of liquid transfers involving volumes of 75 and 100 μL (preset on the pipet). These transfers were executed in both automated (using the system here) and manual (by hand) modes onto a precision weighing scale of 0.001 g resolution. The measurements (Figure 5 and Table 1) revealed no performance disparities between the manual and automated transfer modes. This imputed that the liquid transfer accuracy of the manual pipette was not altered when it was adapted for use in the automated mode (i.e., the system developed here). While gas bubbles during aspiration with the pipette were not found to be present in the tests conducted here, there is always the possibility that they may exist, particularly when handling reagents that comprise surfactants. One precaution that is often exercised is to depress the plunger slowly.¹⁴ One advantage with the system here is the ability to adjust the stepper motor speed and to maintain it consistently (unlike when using it by hand). Another feature that can be included is the incorporation of a wash station to limit the chances of cross-contamination between samples.

It should be noted that with small design changes to this prototype system, virtually any manual pipette that is already available in the laboratory can be adapted to operate as an automated liquid handler in this way. The system also allowed the flexibility of detaching the pipette for manual use if needed. To meet the needs of other more challenging liquid handling tasks, the pipet adapter assembly could be modified instead to fit a multichannel pipette¹⁵ for increased sample throughput processing or a positive displacement pipette¹⁶ if viscous liquids need to be transferred. The material cost for fabricating the system described in this work was only US \$325 (US \$250 for the 3D printer, US \$25 for the stepper motor, and US \$50 for the adapter), thus making this a highly cost attractive option. Minimal effort was also needed to prepare the G-codes to operate the system. Importantly, when the pipetting adapter assembly was detached and the hot-end module was reinstalled, the system was able to resume its operation as a 3D printer. It is also noteworthy that the physical system currently operates with a manual pipette in which the amount of liquid dispensed is determined by the setting made on the pipette before use. It is conceivable that adjustable amounts of liquid can be controlled by altering the stroke of the plunger through more elaborate designs of the actuating mechanisms.

The successful development of this system also presented an opportunity to engage further with a group of students who were part of a pedagogical pilot scheme. It is widely acknowledged that university and high school education faces the critical challenge of how to best support students to transcend their traditional disciplinary boundaries.¹⁷ This is brought about by factors such as changing career aspirations,¹⁸ self-learning challenges,¹⁹ and the impositions of digital technology.²⁰ The participants in the usage exercise cum survey conducted (with the developed system) had just completed a third year (junior) undergraduate teaching unit entitled "Automation Project". This unit was offered to undergraduate students pursuing either Mechanical or Mechatronics Engineering degrees and required them to complete a team-based project to design specific biochemical equipment. This was crafted with the aim of expanding their disciplinary boundaries. The emphasis of the usage exercise and survey carried out here was to engage the 9 participants enlisted to try to link system development to user needs, which was a major component in another of the unit's prime objective of inculcating entrepreneurial mindsets.²¹ When

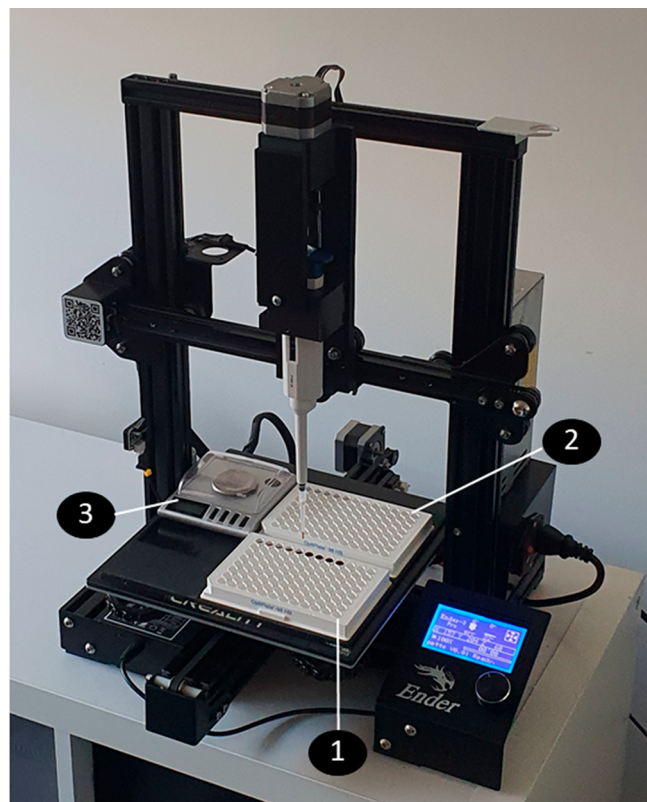


Figure 4. Positional and volumetric performance testing of the prototype automated liquid handler (left panel). Two 96-well microplates, one designated as the source plate containing the liquid to be transferred (1) and the other for liquid receipt (2), were used as well as a precision weighing scale (3) to determine the actual liquid volume transferred.

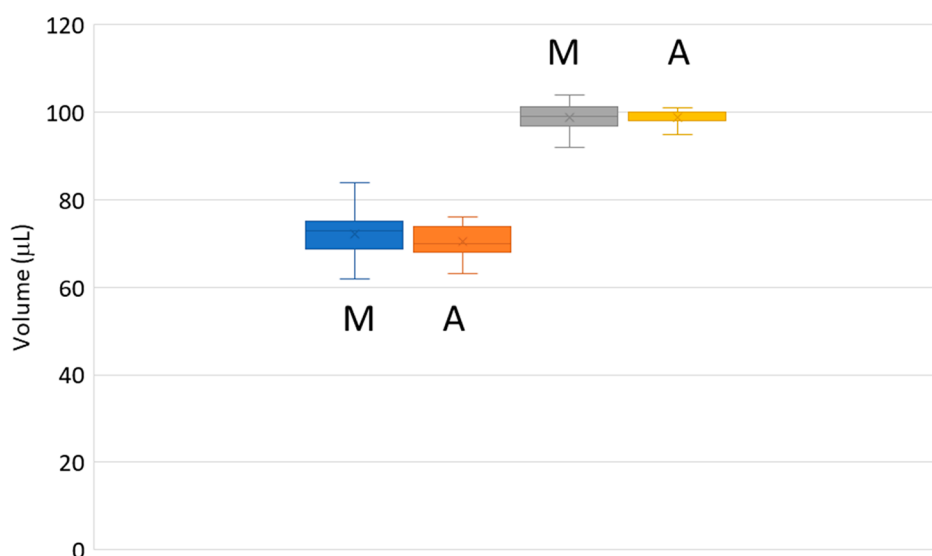


Figure 5. Box plots (right) depict the distribution of volumes transferred (based on 75 and 100 μL settings) when the pipet was used in the manual (M) and automated mode (A) (based on 30 readings each).

Table 1. Data of Liquid Transferred When the Pipette Was Used in the Manual (M) and Automated Mode (A)

Pipette setting = 75 μL		Pipette setting = 100 μL	
Manual	Automated	Manual	Automated
84	75	96	98
76	70	100	100
73	69	97	101
73	67	103	99
75	71	95	100
76	69	95	100
67	74	100	98
74	75	100	100
69	75	104	99
73	64	92	100
72	65	98	98
80	72	99	100
68	68	99	100
77	74	98	101
73	63	99	100
72	73	94	100
73	68	100	96
66	71	100	98
75	70	98	97
70	73	101	100
72	75	99	101
73	75	99	99
71	76	96	100
67	69	95	101
62	74	102	98
71	70	103	97
71	69	97	95
68	69	103	95
75	70	102	100
74	68	101	98
74	69	104	98

presented with the question of which high school level that the exercise was most suited for, the majority of the participants identified Years 10 to 12 (see Figure 6), a finding that concurred with the opinion of an independent educator who

had extensive experience teaching science at all levels to high school students. In a supplementary question posed, all of the participants felt that the exercise could be adapted to serve as a teaching tool for first year undergraduates if software coding requirements were incorporated as part of the usage exercise. Interestingly, some of the participants (without prompting) also indicated that this suitability could be extended beyond those involved in the design and fabrication of instrumentation to those who needed to use instrumentation, as well. The latter, which is imputed in offering computer programming to operate equipment as part of chemistry or biology education coverage at the undergraduate level, offers another way to help in the technology adoption challenge mentioned earlier to nurture interest in adopting laboratory automation usage. It also arguably presents a more attractive way to encourage the use of programming (i.e., writing a series of codes and watching the equipment respond to them) than the current mode of mainly applying it to understand subject concepts better.²²

CONCLUDING REMARKS

The low cost and easily programmable automated liquid handler approach reported here is likely to be a useful learning tool for students in the laboratory. With further development, it may be possible to match the precision of existing commercial automated liquid handlers that cost orders of magnitude more to acquire. If achievable, more of these systems can be made available in university laboratories to facilitate research efforts that require the use of automated liquid handlers. This may help to address the difficulties in realizing do-it-yourself laboratory automation²³ and potentially, with more advanced adaptation, facilitate chemical synthesis work that increasingly relies on the use of robotics.²⁴ It is also conceivable that syringes could take the place of manual pipettes to conduct safe experiments involving gases in an automated manner.²⁵ The 3D printer platform adaptation approach here contrasts with other endeavors in the same vein that necessitate the use of more complicated electronic interfacing methods.^{26,27} It is also anticipated that with a larger cohort of survey participants, the technology acceptance

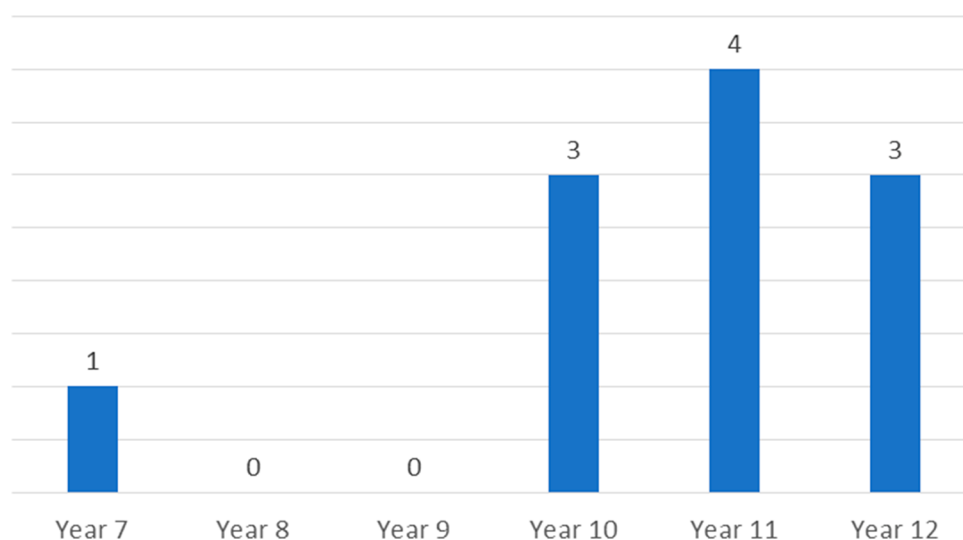


Figure 6. Student responses to the query of which high school level the exercise that they undertook would be best suited for. In the Australian school system, Years 7 to 11 typically equate to ages of 13 to 18 years. The results were based on the input of 9 participants and each student can indicate more than a single year level in their selection.

model could be applied to assess the intent to use and the actual use of the automated liquid handler.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available at <https://pubs.acs.org/doi/10.1021/acs.jchemed.3c00855>.

A short video illustrating the operation of the automated liquid handling system (MP4)

CAD files used to create the adapter assembly using 3D printing (ZIP)

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Notes

The authors declare no competing financial interest.

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